

		ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment 5			
Branch		CSIT		Programme		B.Tech	
Course Name		Full-Stack Web Development with MERN		Semester		6 th	
Course Code		CSE 3839		Academic Year		2025-2026	
Lab Assignment: 5				Topic: Search Engine Optimization			
Learning Level (LL)		L1: Remembering		L3: Applying		L5: Evaluating	
		L2: Understanding		L4: Analysing		L6: Creating	
Q's	Questions					COs	LL
1.	Design and implement a server-side rendered web application using React and Express.js to display event details (seminars, workshops, webinars) . Each event must be accessible via clean URLs, and the system should apply SEO techniques like dynamic meta tags, structured data, sitemap.xml, and robots.txt to improve search engine visibility.					CO1,CO2, CO5	L3
	1. Server-Side Rendering (SSR) • Use Express.js to create routes for event pages. • Render event data using React on the server with <code>ReactDOMServer</code>. • Send fully rendered HTML pages to the client for better SEO and performance.						
	2. Event Pages with Clean URLs • Create routes like <code>/event/:eventName</code> (e.g., <code>/event/react-workshop</code>). • Ensure URLs are readable and descriptive instead of using IDs. • Display event details such as title, date, location, and description.						
	3. Dynamic Meta Tags • Generate dynamic <code><title></code> tags based on event name. • Add meta description using event details (date, topic, location). • Ensure each event page has unique SEO metadata.						
	4. Structured Data (Schema Markup)						

	• Add JSON-LD structured data for events inside <code><script></code> tag. • Use schema type “Event” with properties like name, date, location. • Help search engines understand and display event information properly. <pre>const events = [{ name: "react-workshop", date: "2026-06-10", location: "Kolkata" }, { name: "ai-seminar", date: "2026-07-15", location: "Delhi" }];</pre> 5. Sitemap.xml Generation • Create <code>/sitemap.xml</code> route listing all event URLs. • Dynamically include all available events in XML format. • Help search engines discover and index all event pages. 6. robots.txt Implementation • Create <code>/robots.txt</code> route to control crawler access. • Allow search engines to crawl public event pages. • Optionally block admin or private routes.		
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1. Assignment carries a weightage of 20 marks out of 100
2. CO1,CO2 ,CO5 outcomes are covered.

Course Outcomes	CO1	Understand and configure a modern full-stack development environment using Node.js, MongoDB, Docker, and supporting tools.
	CO2	Design and implement RESTful backend services using Express, MongoDB (Mongoose), and unit testing frameworks.
	CO3	Develop interactive frontend applications using React and efficiently integrate them with backend APIs using modern data-fetching techniques.
	CO4	Apply DevOps practices to containerize, test, and deploy full-stack applications using Docker and CI/CD pipelines.
	CO5	Implement authentication, performance optimization, Search Engine Optimization techniques, and automated testing for scalable and secure web applications.
	CO6	Build and integrate advanced full-stack architectures using data aggregation,

		visualization, and GraphQL APIs with modern frontend clients.
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